

Xavier Theimer-Lienhard

AI Research Engineer - LLM Training & Evaluation

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PROFESSIONAL SUMMARY

AI Research Engineer specializing in large-scale open-weight LLMs for medicine. Led development of 8B and 70B medical models, including multinode training on up to 512 GPUs and distributed evaluation pipelines. Experienced in instruction tuning, reinforcement learning, and benchmark design, with close collaboration with clinicians.

EDUCATION

- **MSc Thesis (Visiting researcher), Yale University** New Haven, US, 2024-2025
- **MSc of Data Science, EPFL**, 5.4/6.0 GPA Lausanne, CH, 2022-2025
- **BSc of Computer Science, EPFL** Lausanne, CH, 2018-2022

EXPERIENCE

AI Research Engineer, EPFL, LiGHT lab Apr 2025-present

Trained Meditron-Apertus, an open source 8B/70B medical LLM trained on a curated medical instruction-tuning dataset. Built and optimized multinode training on HPC clusters (up to 512 GPUs). Designed evaluation harness and medical benchmarks for model comparison.

Research Associate, Yale University Oct 2024-Apr 2025

Trained Meditron-Reasoning, a reasoning medical LLM as part of my master thesis. Improved MMLU accuracy by +30% via synthetic reasoning data and post training. Supervised and coordinated a team of 2-4 students. Led synthetic data generation & reinforcement learning projects.

Research Assistant, EPFL, LiGHT lab Sep 2023-Sep 2024

Core contributor to finetuning of open weight 70B and 8B medical LLMs. Designed large scale medical instruction tuning datasets. Implemented test time scaling pipeline, improving accuracy by 5%.

AI Intern, Irbis Consulting SA Jul 2023-Aug 2023

Trained a Tensorflow neural network captcha solver. Developed an automation app for bidding offers.

PUBLICATIONS & TALKS

Llama-3-Meditron: An Open-Weight Suite of Medical LLMs, AAI 2025 GenAI4Health, [link](#)

Open weight medical LLM built on Llama 3.1

GPoeT: A language Model for Rhyme Generation on Synthetic Data, ACL 2023, [link](#)

Trained a GPT-2 poet that writes text with rhymes.

TECHNICAL SKILLS

Languages: python, pytorch, bash

ML/LLM: transformers, RLHF, DPO, distributed training, evaluation harnesses

Infrastructure: SLURM, HPC clusters, multi-node training, deepspeed, huggingface

Data: Dataset curation, synthetic data generation, benchmarking